

Ayush Kushwaha

+91 7398722981 | ayushkush2402@gmail.com | [LinkedIn](#) | [Kaggle](#) | [GitHub](#)

EXPERIENCE

Assistant Engineer – ML/DL

Jun 2025 – Present

WESEE, Indian Navy (contracted via Softel Solutions Pvt. Ltd.)

New Delhi, India

- Built an end-to-end multimodal, multilingual speech-to-text pipeline using Whisper and NLLB for 8+ languages, reducing transcription latency by 35% through batched inference and beam-search optimization.
- Migrated production RAG vector-search infrastructure from ChromaDB to Milvus, reducing retrieval latency from 120 ms to 67 ms (1.8×) and improving indexing throughput for 10M+ embeddings.
- Deployed speech and transformer-based models with NVIDIA Triton Inference Server, optimizing GPU inference and model serving for production workloads through batching and concurrent request processing.

AI Researcher, ML (Part-Time)

July 2026 – Present

Indian Electoral Lab

New Delhi, India

- Working as quantitative researcher for an early-stage electoral data consulting venture, exploring predictive models on publicly released ECI Form 20 and Form 6 data.
- Designing a tiered model roadmap for booth-level electoral analysis, spanning classification, regression, and causal inference approaches.

Research Intern – Generative Models for SAR Imagery

Jan 2025 – May 2025

DRDO × JNU Joint Project | Certificate

New Delhi, India

- Developed SAR-to-optical unpaired image translation pipeline using CycleGAN with attention-augmented U-Net generators; achieved PSNR >17 dB and SSIM ≈0.65 on DRDO satellite imagery dataset.
- Evaluated diffusion-based image translation pipelines and established production baselines across 3 architectures.
- Reduced per-epoch training time from 18h to 10h (1.8×) via PyTorch DDP, gradient accumulation, and mixed-precision on the PARAM Rudra HPC cluster.

Research Intern – Non-Linearity in Deep Learning

Jun 2024 – Dec 2024

Jawaharlal Nehru University | Certificate

New Delhi, India

- Proposed a novel image augmentation technique targeting non-linearity enhancement in CNNs; validated generalisation improvement via ablation studies on CIFAR-10 across depth, width, and activation variants.
- Implemented distributed training framework achieving 1.7× speedup via PyTorch DDP and automatic mixed-precision on multi-GPU setup; profiled gradient flow to diagnose saturation under proposed augmentation scheme.

PROJECTS

HackMe: Multi-Agent Data Analysis Pipeline | [GitHub](#)

Apr 2026

- Orchestrated 20+ specialised, persona-conditioned AI agents (EDA, planning, modelling teams) for context-aware, autonomous end-to-end analysis of mixed-format datasets using LangGraph and Claude API, with tool use and context management across the pipeline.
- Built a 3-layer memory system (working context, ChromaDB with temporal decay, SQLite knowledge graph) with hybrid BM25 + dense vector retrieval; enabled cross-run continuity and context-aware decision making.

CSIRO Pasture Biomass Prediction | [GitHub](#) | *Kaggle Silver Medal*

Jan 2026

- Achieved $R^2 \approx 0.62$ on a 357-image dataset using a multimodal ensemble of three DINOv2-Huge (632M params each) models with LocalMambaBlock cross-attention fusion; applied dual 512×512 crops and 4-fold CV.
- Designed a “predict 3, derive 2” architecture enforcing mathematical consistency constraints; applied Box-Cox / log1p transforms and managed overfitting via 5-epoch encoder freezing and differential learning rates.

EDUCATION

Jawaharlal Nehru University

New Delhi, India

B.Tech – Electronics and Communications Engineering

2021 – 2025

ACHIEVEMENTS

Kaggle Silver Medal – CSIRO Pasture Biomass Prediction | Certificate

Jan 2026

Open Source – Merged fix into Unsloth (63k★) resolving completion-only loss edge case | PR #4512

Mar 2026

Competitive Programming – Codeforces Rank 64, Round 1080 Div. 3 | Contest Result

Jan 2026

TECHNICAL SKILLS

AI Agents & LLM Systems: Claude API, Multi-Agent Orchestration, Prompt Engineering, Tool Use, Context Management, Hybrid RAG (BM25 + dense), Rapid Prototyping

Computer Vision & Multimodal Learning: CNNs, Vision Transformers, GANs (CycleGAN), Multimodal Fusion, Vision-Language-Audio Pipelines, Context-Aware Modelling

Mathematics & Theory: Probability, Stochastic Processes, Optimisation (SGD/Adam/scheduling), Linear Algebra

Deep Learning: PyTorch, Diffusion Models, NLP/LLMs, Reinforcement Learning (basics), Variational Inference

ML Systems: Distributed Training (DDP, gradient accumulation), HPC (SLURM/PARAM Rudra), MLflow, ONNX

Deployment & Infra: FastAPI, Docker, Milvus, ChromaDB, Jetson Orin (edge), LangChain, LangGraph

Languages: Python (primary), C++ (competitive programming), SQL